



FACE-SWAP DEEPFAKES DETECTION USING NOVEL MULTI-DIRECTIONAL HEXADECIMAL FEATURE DESCRIPTOR

PRESENTER

QURAT-UL-AIN, PHD SCHOLAR(UET TAXILA)

RESEARCH ASSISTANT

MSP LAB

SOFTWARE ENGINEERING DEPARTMENT

UNIVERSITY OF ENGINEERING AND TECHNOLOGY-TAXILA, PAKISTAN

16th August 2022

19th IBCAST 2022



Multimedia Signal Processing Lab

2

MSP research lab focuses on generating cutting-edge research results in the field of multimedia signal processing. MSP is conducting useful research and designs various state-of-the-art computer vision, image and audio processing based applications by using both learning and non-learning-based techniques. The specific application areas associated with research in our lab include:

- Audio Processing: Audio stream analysis for excitement detection, speech and music segregation, speech recognition, audio forensics.
- Video Processing: Video summarization, DeepFakes, surveillance, video analytics, understanding, segmentation, object tracking and recognition, content-based retrieval, activity and gesture recognition, multimedia applications.
- Image Processing: Brain Aneurysm Detection and Rupture Prediction, Morphological image processing, tracking and recognition, segmentation, computer vision, image processing for biometrics.



3

Presentation Outline

- ❑ Introduction (Background and Motivation)
- ❑ Problem Statement (Challenges in state-of-art descriptors)
- ❑ Aims and Objectives
- ❑ Literature Work
- ❑ Contributions
- ❑ Proposed Methodology(face-swap deepfakes deketection)
- ❑ Performance Evaluation
- ❑ Conclusion



Introduction

4

- ❑ The growing number of sophisticated deep learning algorithms lead in creation of highly realistic deepfake videos.
- ❑ face-swap is the most commonly employed identity swapping deepfakes approach.
- ❑ The motivation is to develop a robust local texture descriptor to extract more directional and magnitude details from face-swap deepfake detection.



Fig.1:Original frames



Fig.2: GAN generated face-swap frames



Problem Statement: Challenges in Face-swap Deepfake Detection

- Existing descriptors are failed to detect videos containing variations of the facial skin tone of people having different races, illumination conditions, presence of accessories like glasses on the face, and loss of details due to compressed video resolution.
- Methods based on local texture descriptors [17] like LBP, SURF, and LTP compute only limited directional information and disregard the magnitude information.



Fig.3: Frames from datasets containing people of different skin color, illumination conditions, and faces with glasses.



Aims and Objectives

6

- ❑ The major objectives of the proposed research work are:
 - ✓ Our motivation was to develop a robust descriptor by capturing more directional and higher-magnitude details from the adjacent pixels to effectively represent the frames of real and face-swap videos.
 - ✓ A multi-directional feature descriptor, which effectively captures the texture orientation and magnitude information from the video frames.
 - ✓ A deepfakes detection system that is robust on videos containing variations of the facial skin tone of people having different races, and illumination conditions.



Problems in Existing Literature

7

Table 1. LITERATURE REVIEW OF CONTEMPORARY METHODS.

Methods	Technique	Problems
[9]	SVM Classifier+ landmarks	Fail to detect blur images.
[12]	SVM classifier	Degrade performance on person off-looking frames.
[15]	Texture energy-based features + (MLP, LogReg)	Fail to detect faces with closed eyes.
[8]	SURF + SVM	Not robust on unseen data.
[13]	EAR+ landmarks	Fail to detect video frame with more eye blinking scenarios.
[21]	Deep Features + CNN	Not robust on detection of unseen videos.
[22]	Deep Features + SVM	Performance fails on compressed videos.



Contributions

8

The main contributions of the proposed work are:

- ✓ We propose a novel multi-directional hexadecimal feature descriptor (MDHFD) to effectively capture the texture orientation and magnitude information from the video frames.
- ✓ Our proposed face-swap deepfakes detection system is robust on videos containing variations of the facial skin tone of people having different races, illumination conditions, presence of accessories like glasses on the face, and loss of details due to compressed video resolution.
- ✓ Rigorous experimentation was performed on two diverse deepfakes datasets including the cross-corpora evaluation to test the generalizability of our method.

Face-swap Deepfakes Detection Framework

9

Multi-directional hexadecimal feature descriptor

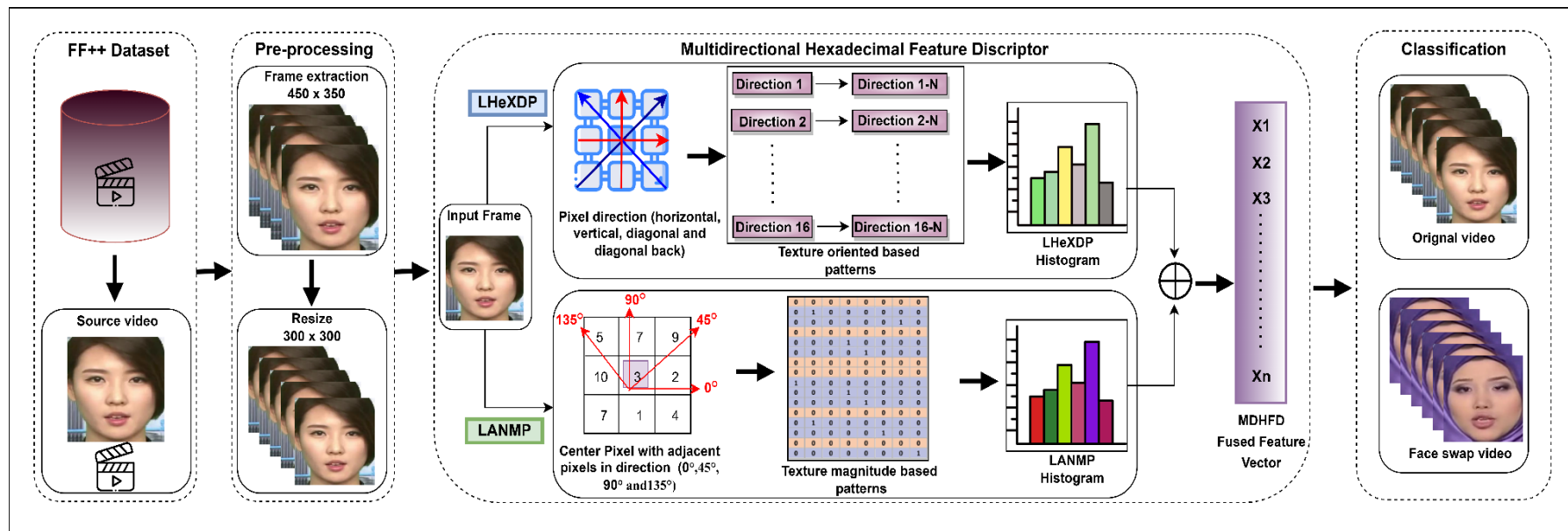


Fig.4: Proposed method

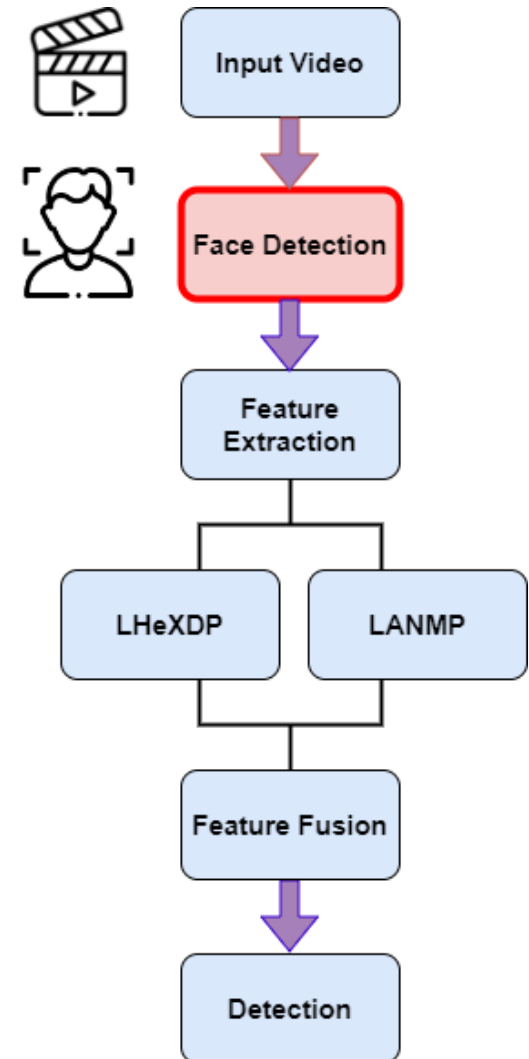


Face-swap Deepfakes Detection Framework

10

❑ Face Detection

- ✓ Multi-task Cascaded Convolutional Networks (MTCNN) [18] is used for frame-level extraction.
- ✓ MTCNN detect and extract only the face portion from the input video.
- ✓ In comparison with other face detectors like Haar Cascade, MTCNN detects the faces precisely in the presence of occlusion and varying illumination conditions.





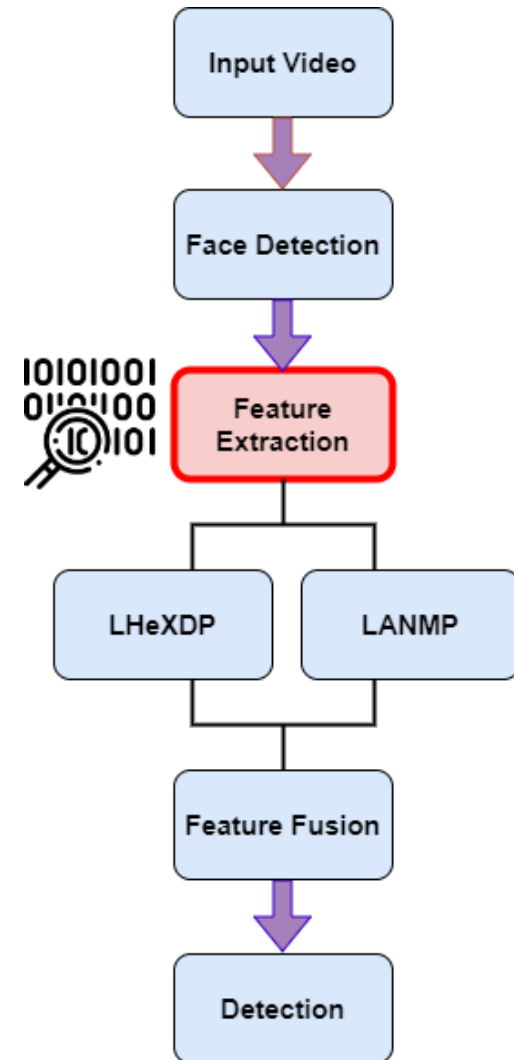
Face-swap Deepfakes Detection Framework

11

Feature extraction

▪ *Multi-directional hexadecimal feature descriptor*

- ✓ The proposed descriptor is comprised of the direction and magnitude-based features.
- ✓ The orientation patterns effectively extract discriminative information from the frames and compute directional information.
- ✓ Furthermore, we compute the patterns based on magnitude which captures additional useful information.
- ✓ Each resultant feature vector is a combination of magnitude and orientation patterns.





Face-swap Deepfakes Detection Framework

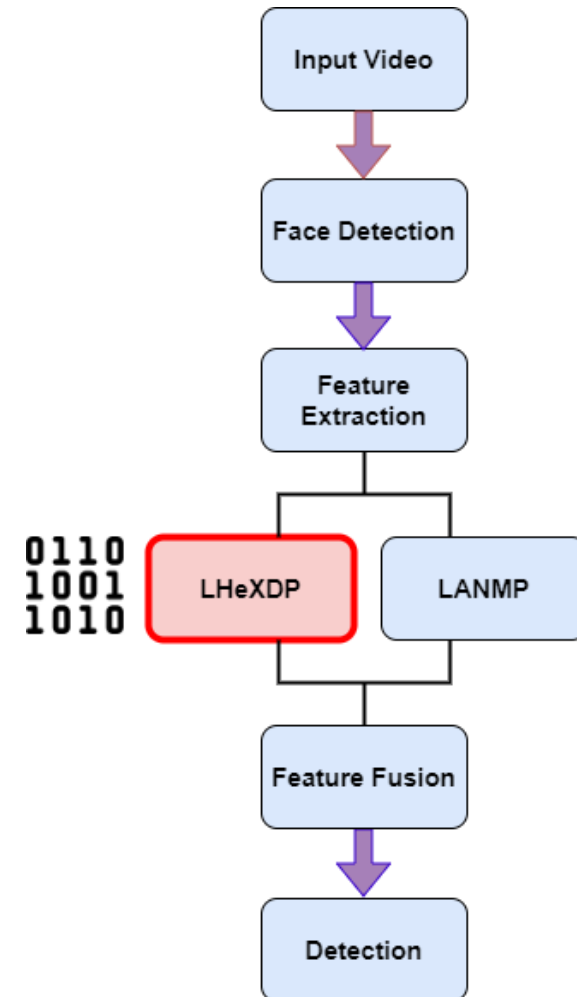
12

□ LHeXDP

- ✓ For directional information, we employed a local hexadecimal feature descriptor (LHeXDP).
- ✓ For a given frame $F(x, y)$, we compute the 1st order derivative at the grayscale value of the surrounding pixels along with directions, as shown in (1).

$$F_{\alpha}^1(d_{h,d,v,db})|\alpha = 0^0, 45^0, 90^0, 135^0 \quad (1)$$

- ✓ The frame is translated into 16 different values from which the directions are determined.
- ✓ We construct texture orientation for each pixel by comparing the 1st order derivatives of the center pixel direction with the directions of all the eight surrounding neighbors.





Face-swap Deepfakes Detection Framework

13

□ LHeXDP

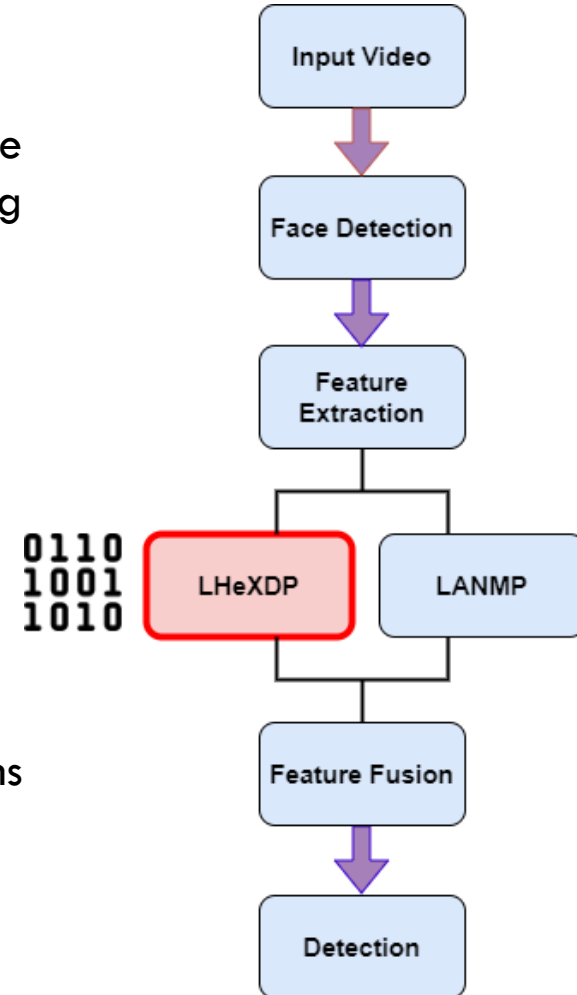
- ✓ Taking the 2nd order derivative of the central pixel, we obtained 8-bit directions with all the eight neighbouring directions ($s=1-8$) using the (2) and (3).

$$LHXDP^2 = \left\{ \begin{array}{l} T_1 \left(F_{dir}^1(d_c) \cdot F_{dir}^1(d_1) \right) \cdot T_1 \left(F_{dir}^1(d_c) \cdot F_{dir}^1(d_2) \right) \dots \\ \dots, T_1 \left(F_{dir}^1(d_c) \cdot F_{dir}^1(d_s) \right) |_{s=8} \end{array} \right\} \quad (2)$$

$$F_{dir}^1(d_c) \times F_{dir}^1(d_s) = \begin{cases} 0, & F_{dir}^1(d_c) = F_{dir}^1(d_s) \\ F_{dir}^1(d_s), & \text{otherwise} \end{cases} \quad (3)$$

- ✓ The 2nd order is distributed into 16 binary patterns computed as shown in (4).

$$\begin{aligned} & LHXDP^2 |_{directions=1,2,3,4,5,6,7,8,9,10,11,12,13,15,16} \\ &= \sum_{s=0}^s 2^{(s-1)} \times T_1(LHXDP^2(d_c)) |_{directions=\alpha} \end{aligned} \quad (4)$$





Face-swap Deepfakes Detection Framework

14

LANMP

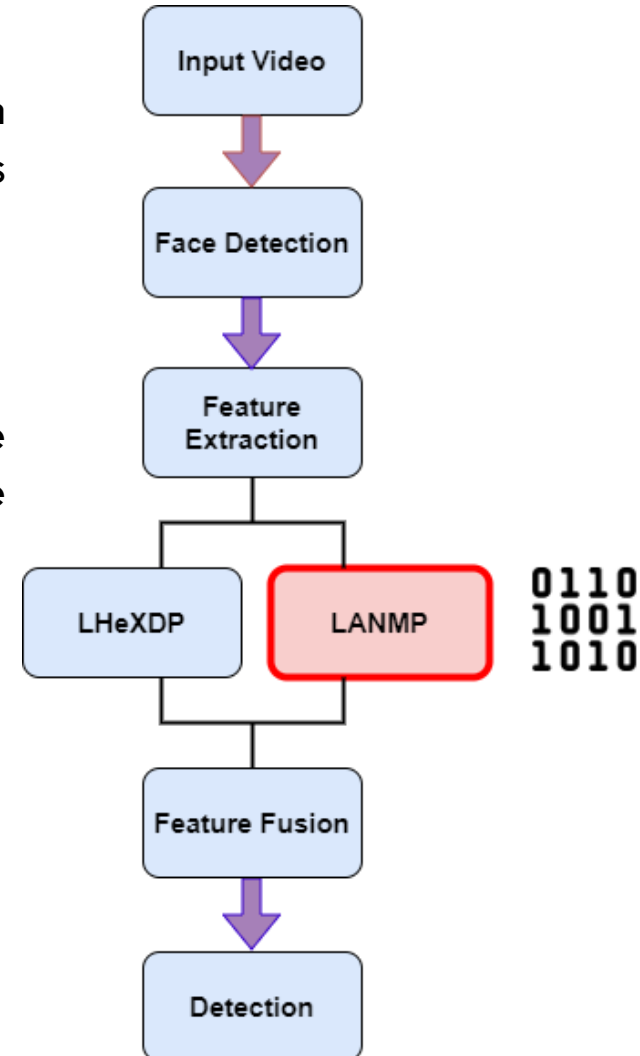
- ✓ Local Adjacent Neighborhood Magnitude Pattern (LANMP) compute 1st order derivatives that captures edge information of each frame in detailed.

$$M_1^1(d_s) = \sqrt{(F_{0^0}^1(d_s))^2 + (F_{45^0}^1(d_s))^2 + (F_{105^0}^1(d_s))^2 + (F_{135^0}^1(d_s))^2} \quad (5)$$

- ✓ In (7), R is the function that discriminates between the magnitude of surrounding neighbors' pixels and the center pixel based on parameter x .

$$LANMP^2 = \sum_{s=1}^S 2^{(s-1)} \times R(M_1^1(d_s) - M_1^1(d_c))|_{s=8} \quad (6)$$

$$R(x) = \begin{cases} 1, & x \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (7)$$





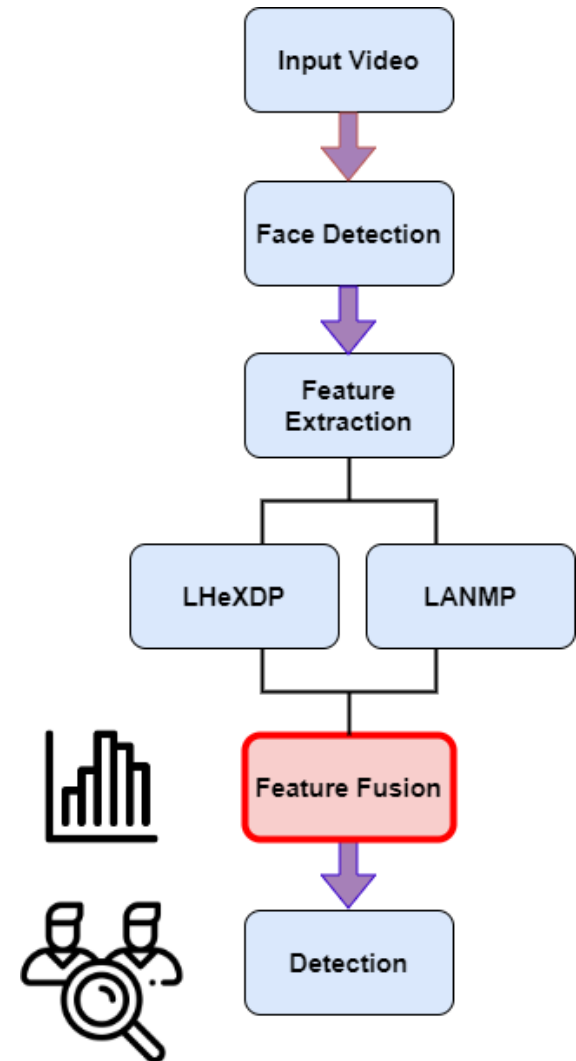
Face-swap Deepfakes Detection Framework

15

Feature Fusion

- ✓ After extracting the local patterns (LHeXDP and LANMP) of each frame, we obtain the histogram of both patterns.
- ✓ The proposed MDHDF is formed by fused histograms of LHXDP and LANMP.

$$\text{Hist}^{MDHFD} = [\text{Hist}^{LHXDP} || \text{Hist}^{LANMP}] \quad (8)$$



Performance Evaluation

16

DATASETS

We evaluated the performance of the proposed method on the original and face-swapped subset of following two datasets.

- Face Forensics++ dataset (FF++) [22].
- World Leaders dataset (WLDR) [12].

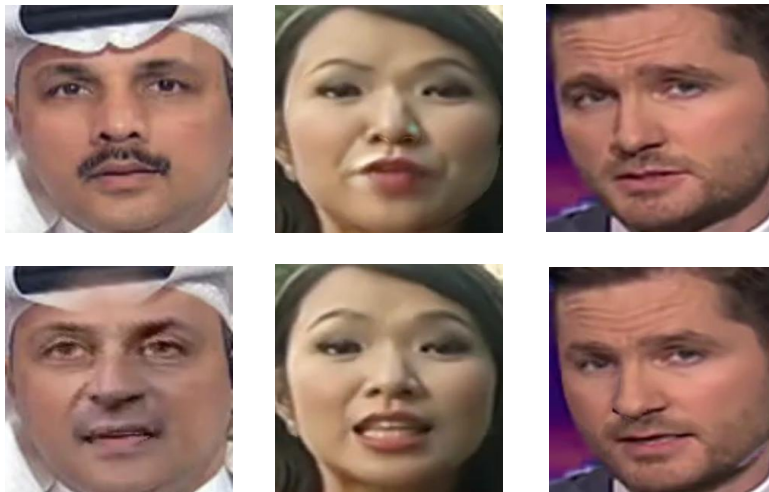


Fig.5: Samples from FF++ Dataset,
Top row contain original and bottom row contain Faceswap sample.



Fig.6: Samples from WLDR Dataset,
Top row contain original and bottom row contain Faceswap sample.



Performance Evaluation

17

- ❑ Performance Evaluation of proposed method
- ✓ To evaluate the effectiveness of our method for face-swap deepfakes detection, we designed an experiment to evaluate the performance of our method on the original and face swap subset on both the FF++ and the WLDR datasets.
- ✓ We resized all frames to 300×300 and extracted the features using our MDHFD for both the training and testing set.
- ✓ These features are then used to train the SVM classifier for video deepfakes detection. SVM produce a better result on proposed MDHFD features using a gaussian kernel.
- ✓ More specifically, we achieved an accuracy of 92.3% and AUC of 0.99 on the FF++ dataset, whereas achieved an accuracy of 96.9% and AUC of 1.00 on the WLDR dataset.



Performance Evaluation

18

- ❑ Confusion matrix Analysis
- ✓ To better investigate the false-acceptance and rejection scenarios, we designed a confusion matrix analysis to depict the classification performance of our method on both datasets as shown below.

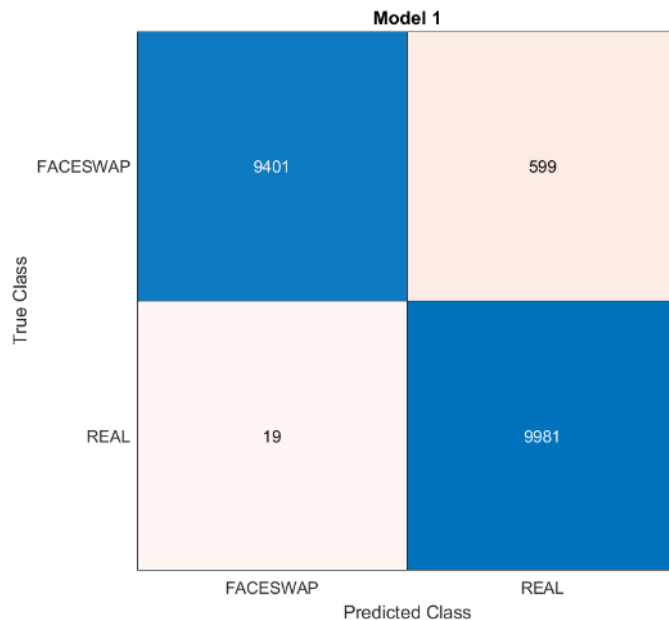


Fig.7. Confusion Matrix of a Proposed Method on WLDR dataset.

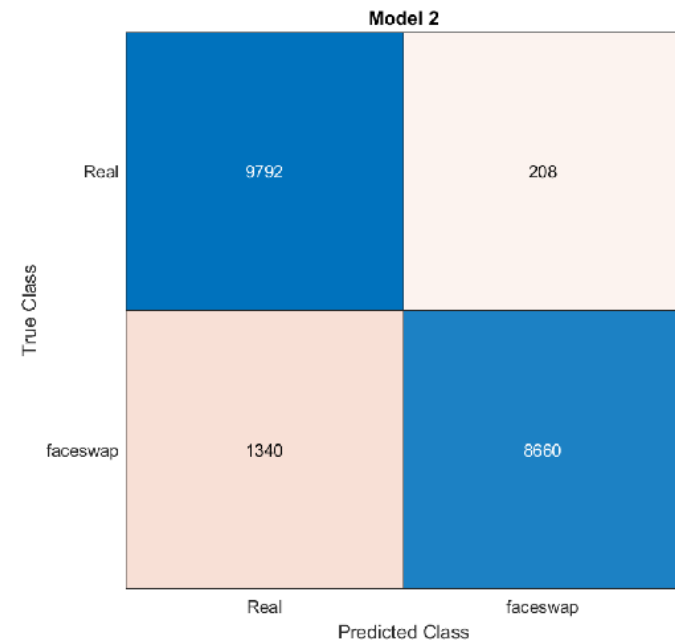


Fig.8. Confusion Matrix of a Proposed Method on FF++ dataset.

Performance Evaluation

19

- ❑ ROC Curve Analysis
- ✓ ROC curve analysis is performed in our third experiment to illustrate the performance of the proposed method for face-swap deepfake detection.
- ✓ Using SVM with proposed features we achieved the best AUC of 0.99 on the FF++ dataset. On the WLDR dataset, we achieved the best AUC of 1.00 with SVM.

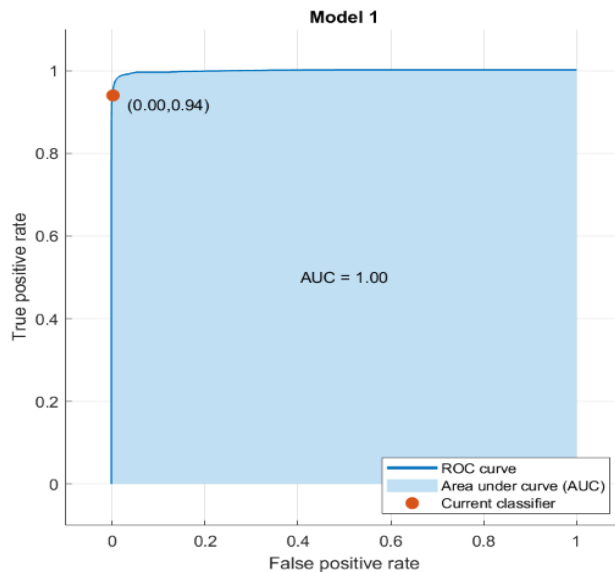


Fig.9. ROC of a Proposed Method on WLDR dataset.

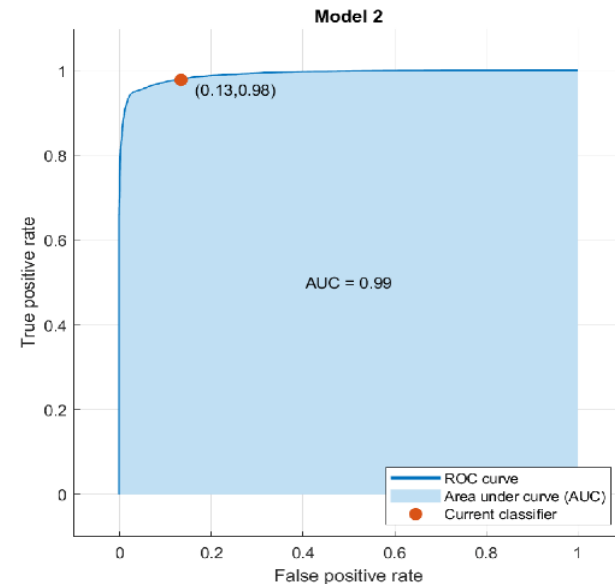


Fig.10. ROC of a Proposed Method on FF++ dataset.



Performance Evaluation

20

- Performance comparison with other classifiers

Table 2. PERFORMANCE EVALUATION OF PROPOSED METHOD ON DATASETS USING DIFFERENT CLASSIFIERS.

Classifiers	FF++ Dataset		WLDR Dataset	
	Accuracy	AUC	Accuracy	AUC
SVM	92.3%	0.99	96.9%	1.00
Bagged Trees Ensemble	89.9%	0.97	95.6%	1.00
Narrow Neural Network	85.49%	0.94	95.5%	1.00
Fine Trees	77.3%	0.82	91.9%	0.98
KNN	80.4%	0.92	94.5%	0.99



Performance Evaluation

21

❑ Cross-Corpora Evaluation

- ✓ To test the generalizability of our method, we designed a two-stage cross-corpora evaluation experiment using the proposed features.
- ✓ In the first stage of this experiment, we used WLDR dataset to train our model and evaluated it on the FF++ dataset. We achieved 98.8% training accuracy but attained a less accurate result of 49.6% and an AUC of 0.69 on the test set.
- ✓ In the second stage of this experiment, we used the FF++ dataset for training and the WLDR dataset for testing and achieved an accuracy of 93.4% on training but attained less accurate results of 50% accuracy and 0.50 AUC on the test set.
- ✓ Cross-corpora experiments attained lower results because both FF++ and WLDR dataset are different from each other. For instance, videos are diverse in terms of illumination and occlusion conditions, presence of accessories like glasses on the face, and loss of details due to compressed video resolution.



Performance Evaluation

22

- Performance comparison with contemporary methods.

Table 3. PERFORMANCE EVALUATION OF PROPOSED METHOD WITH CONTEMPORARY METHODS.

	Methods	Technique	Results
AUC	Proposed Method	MDHFD + SVM (FF++)	0.99
		MDHFD + SVM (WLDR)	1.00
	[9]	SVM Classifier+ landmarks	0.89
	[10]	SVM+ Multimedia stream descriptor	0.93
	[12]	SVM classifier	0.96
	[15]	texture energy-based features + (MLP, LogReg)	0.851 0.784
	[16]	Landmarks+(VGG16, ResNet50, ResNet101, ResNet152)	0.84 0.97 0.95 0.93
Accuracy	Proposed Method	MDHFD + SVM (FF++)	92.3%
		MDHFD + SVM (WLDR)	96.9%
	[8]	SURF + SVM	92%
	[13]	EAR+ landmarks	87.5%
	[21]	Deep Features + CNN	83.71%
	[22]	Deep Features + SVM	90.29%



Conclusion

23

- In this research work, we have presented a novel texture feature descriptor to reliably capture the orientation and magnitude-oriented details from the input video frames that are then used to detect the face swap deepfakes.
- The proposed method better addresses the problem of face swap-based deepfakes detection under challenging conditions such as variations in the face skin tone of people having different races, illumination conditions, presence of accessories like glasses on the face, etc.
- We evaluated the performance of the proposed method on the face-swap subset of Face Forensics++ and the WLDR dataset that contains all the aforementioned challenges.
- Experimental results on both datasets illustrate the effectiveness of the proposed method over the state-of-the-art methods for better detection of face-swap deepfakes.
- In the future, we plan to extend our work to detect multiple types of deepfakes and will try to improve our results on cross-corpora evaluation.



References

- [1] FakeApp 2.2.0, Available at: <https://www.malavida.com/en/soft/fakeapp/>. Accessed: September 18, 2020.
- [2] Faceswap: Deepfakes software for all, Available at: <https://github.com/deepfakes/faceswap>. Accessed: September 08, 2020.
- [3] G. Antipov, M. Baccouche, and J.-L. Dugelay. Face aging with conditional generative adversarial networks. arXiv:1702.01983, Feb. 2017.
- [4] A. Tewari et al. Mofa: Model-based deep convolutional face autoencoder for unsupervised monocular reconstruction. Proceedings of the IEEE International Conference on Computer Vision Workshops, pages 1274–1283, Oct. 2017. Venice, Italy. [
- [5] J. F. Boylan, "Will deep-fake technology destroy democracy?" The New York Times, Oct, vol. 17, 2018
- [6] A. Khodabakhsh, R. Ramachandra, K. Raja, P. Wasnik, C. Busch, in 2018 International Conference of the Biometrics Special Interest Group (BIOSIG). Fake face detection methods: can they be generalized? (IEEE, 2018). <https://doi.org/10.23919/biosig.2018.8553251>.
- [7] Y. Zhang, L. Zheng, and V. L. Thing, "Automated face swapping and its detection," in 2017 IEEE 2nd International Conference on Signal and Image Processing (ICSIP), 2017, pp. 15-19: IEEE.
- [8] A. Agarwal, R. Singh, M. Vatsa, A. Noore, in 2017 IEEE International Joint Conference on Biometrics (IJCB). Swapped! Digital face presentation attack detection via weighted local magnitude pattern (IEEE, 2017).
- [9] X. Yang, Y. Li, and S. Lyu, "Exposing deep fakes using inconsistent head poses," in ICASSP 2019- 2019 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2019, pp. 8261-8265: IEEE.
- [10] D. Güera, S. Baireddy, P. Bestagini, S. Tubaro, and E. J. Delp, "We Need No Pixels: Video Manipulation Detection Using Stream Descriptors," arXiv preprint arXiv:1906.08743, 2019.
- [11] K. Jack, "Chapter 13-MPEG-2," Video Demystified: A Handbook for the Digital Engineer, pp. 577- 737.
- [12] Shruti Agarwal, Hany Farid, Yuming Gu, Mingming He, Koki Nagano, and Hao Li. Protecting world leaders against deep fakes. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, pages 38–45, 2019.
- [13] T. Jung, S. Kim, and K. Kim, "DeepVision: Deepfakes Detection Using Human Eye Blinking Pattern," IEEE Access, vol. 8, pp. 83144- 83154, 2020.
- [14] T. Soukupova and J. Cech, "Eye blink detection using facial landmarks," in 21st computer vision winter workshop, Rimske Toplice, Slovenia, 2016.
- [15] F. Matern, C. Riess, and M. Stamminger, "Exploiting visual artifacts to expose deepfakes and face manipulations," in 2019 IEEE Winter Applications of Computer Vision Workshops (WACVW), 2019, pp. 83-92: IEEE.
- [16] Qurat-ul-ain, N. Nida, A. Irtaza, and N. Ilyas, "Forged Face Detection using ELA and Deep Learning Techniques," 2021 International Bhurban Conference on Applied Sciences and Technologies (IBCAST), 2021, pp. 271-275, doi: 10.1109/IBCAST51254.2021.9393234.
- [17] Akhtar, Z., & Dasgupta, D. (2019, November). A comparative evaluation of local feature descriptors for deepfakes detection. In 2019 IEEE International Symposium on Technologies for Homeland Security (HST) (pp. 1-5). IEEE.
- [18] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," IEEE Signal Processing Letters, vol. 23, no. 10, pp. 1499-1503, 2016.
- [19] P. Korshunov, S. Marcel, Deepfakes: a new threat to face recognition? assessment and detection. arXiv preprint arXiv:1812.08685 (2018).
- [20] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, "Mesonet: a compact facial video forgery detection network," in 2018 IEEE International Workshop on Information Forensics and Security (WIFS), 2018, pp. 1-7: IEEE.
- [21] H. H. Nguyen, F. Fang, J. Yamagishi, and I. Echizen, "Multi-task learning for detecting and segmenting manipulated facial images and videos," in 2019 IEEE 10th International Conference on Biometrics Theory, Applications and Systems (BTAS), 2019, pp. 1-8.
- [22] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "Faceforensics++: Learning to detect manipulated facial images," in Proceedings of the IEEE International Conference on Computer Vision, 2019, pp. 1-11.
- [23] Masood, M., Nawaz, M., Malik, K. M., Javed, A., Irtaza, A., & Malik, H. (2022). Deepfakes Generation and Detection: State-of-the-art, open challenges, countermeasures, and way forward. Applied Intelligence, 1-53



For any query Feel Free to ask

